

Stack Problem Bank

Legend

★ = Core Placement Problem (Mandatory)

Language-Agnostic DSA Problem Bank | Implement Stack from scratch using Array and Linked List before using built-ins.

Section 1: Stack Internals — Implementation

★ **Implement Stack using Array (Push, Pop, Peek, isEmpty, isFull)**

★ **Implement Stack using Linked List (Push, Pop, Peek, isEmpty)**

Implement Stack with Dynamic Resizing

★ **Display All Elements of a Stack (Traverse without Destroying)**

Implement Two Stacks in a Single Array

Section 2: Basic Stack Operations

★ **Push N Elements and Pop One by One**

★ **Reverse a String using a Stack**

★ **Check if a Stack is Empty or Full**

Find the Minimum Element in a Stack (using Extra Stack)

★ **Implement a Stack that Supports getMin() in O(1)**

Sort a Stack using Recursion

Sort a Stack using Another Stack

Section 3: Parenthesis and Expression Problems

★ **Check for Balanced Parentheses { [()] }**

★ **Evaluate a Postfix Expression**

★ **Convert Infix Expression to Postfix**

Convert Infix Expression to Prefix

Evaluate a Prefix Expression

★ **Convert Postfix to Infix Expression**

Check Redundant Brackets in an Expression

Section 4: Next Greater / Smaller Element Problems

★ **Next Greater Element to the Right**

★ **Next Greater Element to the Left**

Next Smaller Element to the Right

Next Smaller Element to the Left

★ **Previous Greater Element**

Count Elements Greater than All Elements to the Right

Find Leaders in an Array using Stack

Section 5: Monotonic Stack Problems

★ **Stock Span Problem**

★ **Largest Rectangle in a Histogram**

Maximal Rectangle in a Binary Matrix

★ **Trapping Rain Water using Stack**

Remove K Digits to Get Smallest Number

Online Stock Span Problem

Section 6: Stack-Based Design Problems

★ **Design a Browser History (Back / Forward) using Stack**

★ **Undo-Redo Mechanism using Two Stacks**

Implement Queue using Two Stacks

★ **Celebrity Problem using Stack**

The Car Fleet Problem

Section 7: Real-World Applications of Stack

★ **Function Call Stack Simulation — Trace Recursive Calls with Stack**

★ **Balanced HTML / XML Tag Validator using Stack**

★ **Compiler Symbol Table using Stack (Track Variable Scope)**

Text Editor — Implement Ctrl+Z (Undo) and Ctrl+Y (Redo) using Stack

★ **Syntax Checker for Arithmetic Expressions using Stack**

Backtracking in a Maze — Use Stack to Track Path

Postfix Calculator — Build a Command-Line Calculator using Stack